26-11-15

Indian Institute of Engineering Science & Technology, Shibpur

B.Tech-M.Tech Dual Degree 3^{rd} Semester (CST) Examination, 2015

Digital Logic (CS 301)

Answer any 7 questions.

F.M. 70 Time: 3 hrs

1. (a) Compare 1's and 2's complements.

(b) Obtain the 10's complement of 90090 and 2's complement of 5.

c) Simplify the following Boolean function

$$F(w,\ x,\ y,\ z) = \Sigma(4,\ 5,\ 7,\ 12,\ 14,\ 15) + d(3, 8, 10).$$

[3 + 2 + 5]

2. Draw the circuit diagram of a two inputs DTL NAND gate. Connect the output of the DTL gate to N inputs of other similar gates. Assume that the output transistor is saturated and its base current is $0.44mA$. Let $h_{FE} = 20$. Find the value of N that will keep the transistor in saturation and what is the fan-out of the gate? [3 + 7]

3. Obtain the logic diagram that converts a four-digit binary number to a decimal number in BCD. Note that two decimal digits are needed since the binary numbers range from 0 to 15. [10]

4. (a) Design a combinational circuit that generates the 9's complement of a BCD digit.
(b) Implement the following function with a multiplexer

$$F(A,\ B,\ B,\ D) = \Sigma(0,\ 1,\ 3,\ 4,\ 8,\ 9,\ 15)$$

[5 + 5]

5. (a) Define Moore machine and Mealy machine.

b) A full-adder receives two external inputs X and Y; the third input Z comes from the output of a D flip-flop. The carry output is transferred to the flip-flop in every clock pulse. The external output S gives the sum of X, Y and Z. Obtain the state table and state diagram of the sequential circuit. [3 + 7]

6. (a) What do you mean by lock-out in respect of counter?

(b) Design a Mod-5 synchronous counter using J-K flip-flops so that if at any time the unused states 101, 110 and 111 appear, the next clock will reset the counter to 000. [2 + 8]

7. (a) Why we do not use S = 1 and R = 1 as an input sequence in case of R-S (using NOR gates) flip-flop? What is the problem in J-K flip-flop and how is it removed?

(b) Design a counter with the following binary sequence: 0, 1, 3, 2, 6, 4, 5, 7 and repeat. Use *RS* flip-flops.

[4 + 6]

8. (a) The content of a 4-bit shift register is initially 1101. The register is shifted six times to the right with the serial input being 101101. What is the content of the register after each shift?

(b) Draw the diagram of a 4-bit ripple counter using flip-flop that trigger on the positive edge.

(c) Design a sequential circuit with JK flip-flops to satisfy the following state equations:

$$A(t+1) = xAB + y\overline{A}C + xy$$
$$B(t+1) = xAC + \overline{y}B\overline{C}$$
$$C(t+1) = \overline{x}B + yA\overline{B}$$

[2 + 2 + 6]

9. Write short notes on the following.

[5 + 5]

(i) Bi-direction shift register.
(ii) Tri-state TTL inverter.